

## Topic 11

# Organic chemistry

**Concept guide · Worked examples · Cambridge-style practice · Fully explained answers**

**What this guide teaches**

A complete explanation of the syllabus concepts, chemical reasoning, worked applications, common misconceptions and original exam-style practice for this topic.

**Syllabus map**

|             |                                |
|-------------|--------------------------------|
| <b>11.1</b> | Formulae, names and structures |
| <b>11.2</b> | Fuels and petroleum            |
| <b>11.3</b> | Homologous series              |
| <b>11.4</b> | Alkanes                        |
| <b>11.5</b> | Alkenes                        |
| <b>11.6</b> | Alcohols                       |
| <b>11.7</b> | Carboxylic acids               |
| <b>11.8</b> | Polymers                       |

How to study: retrieve definitions and equations, cover the worked examples, then complete the practice paper without using the notes.

# 1. Representing organic compounds

A molecular formula gives numbers of each atom; a displayed formula shows every atom and bond; a structural formula shows connectivity in a compact form. Members of a homologous series share a functional group and general formula, have similar chemical properties, show gradual physical-property trends and differ successively by  $\text{CH}_2$ . Isomers share a molecular formula but have different structures.

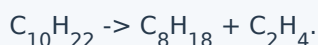
## Examiner insight

Name the longest carbon chain and identify the functional group or multiple bond.

# 2. Petroleum, fuels and cracking

Petroleum is a mixture of hydrocarbons separated by fractional distillation. Fractions condense over temperature ranges; smaller molecules have lower boiling points, are more volatile and ignite more easily. Cracking uses heat and a catalyst to convert long-chain alkanes into shorter alkanes and alkenes, matching demand and supplying monomers.

## Worked example



## Examiner insight

Fractional distillation is physical separation; cracking is a chemical reaction.

# 3. Alkanes and combustion

Alkanes are saturated hydrocarbons with general formula  $\text{C}_n\text{H}_{2n+2}$ . Complete combustion forms carbon dioxide and water; incomplete combustion forms carbon monoxide and/or carbon. Alkanes undergo substitution with chlorine in ultraviolet light, replacing hydrogen by chlorine.

## Worked example



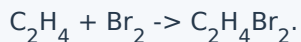
## Examiner insight

“Saturated” means carbon-carbon single bonds only, not that the substance contains water.

## 4. Alkenes

Alkenes are unsaturated hydrocarbons containing C=C and have general formula  $C_nH_{2n}$ . They undergo addition reactions. Bromine water changes orange-brown to colourless, distinguishing an alkene from an alkane. Hydrogen forms an alkane; steam forms an alcohol.

### Worked example



### Examiner insight

Addition uses the double bond and produces one saturated product.

## 5. Alcohols and carboxylic acids

Ethanol can be made by fermentation of glucose using yeast at about 30-40 °C without oxygen, or by hydration of ethene using steam and a catalyst. Alcohols combust and can oxidise to carboxylic acids. Ethanoic acid reacts as an acid and esterifies with ethanol to form ethyl ethanoate and water.

### Worked example

glucose  $\rightarrow$  2ethanol + 2carbon dioxide; ethanol + ethanoic acid  $\rightarrow$  ethyl ethanoate + water.

### Examiner insight

Compare fermentation, which is renewable, dilute and slow, with hydration, which is fast, continuous and uses a fossil feedstock.

## 6. Polymers

Addition polymerisation joins many alkene monomers and forms only the polymer; the C=C becomes C-C in the repeat unit. Condensation polymerisation joins bifunctional monomers and eliminates a small molecule such as water. Polyesters contain ester links; proteins are natural condensation polymers of amino acids.

### Examiner insight

Keep continuation bonds through the brackets of a polymer repeat unit and include the subscript n.

# Exam technique

## High-value habits

Use precise chemical vocabulary, balance equations, show every calculation step, include units and state symbols where required, distinguish observations from conclusions, and answer the command word directly.

## Retrieval checklist

- Recall all definitions, tests and equations without notes.
- Explain one unfamiliar observation using particles, bonding or collision theory.
- Check charge, atoms, units and significant figures in every calculation.
- Complete one timed set and mark it only after finishing.

# Practice Paper

## Topic 11 · Organic chemistry

### Instructions

Answer all questions. Select one option for multiple choice. Show working in open spaces and write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

**Total marks: 50**

Suggested time: 60 minutes.

## Section A · Multiple choice

Choose the best answer for each question.

1. Successive members of a homologous series differ by [1]
  - A CH
  - B CH<sub>2</sub>
  - C C<sub>2</sub>H<sub>2</sub>
  - D OH
  
2. Which is an alkane? [1]
  - A C<sub>3</sub>H<sub>6</sub>
  - B C<sub>4</sub>H<sub>8</sub>
  - C C<sub>4</sub>H<sub>10</sub>
  - D C<sub>2</sub>H<sub>4</sub>
  
3. Which property increases with hydrocarbon chain length? [1]
  - A volatility
  - B ease of ignition
  - C boiling point
  - D flammability
  
4. Cracking produces [1]
  - A only long alkanes
  - B shorter alkanes and alkenes
  - C metals and oxygen
  - D alcohols only
  
5. Complete combustion of ethanol forms [1]
  - A CO and H<sub>2</sub>
  - B CO<sub>2</sub> and H<sub>2</sub>O
  - C C and H<sub>2</sub>
  - D CH<sub>4</sub> and O<sub>2</sub>
  
6. Bromine water is decolourised by [1]
  - A ethane
  - B propane
  - C ethene
  - D methane
  
7. Hydration of ethene produces [1]
  - A ethanol
  - B ethanoic acid
  - C ethyl ethanoate
  - D poly(ethene)

8. Fermentation requires [1]
- A oxygen and 100 °C
  - B yeast and absence of oxygen
  - C a metal catalyst at high pressure only
  - D chlorine and ultraviolet
9. Ethanol oxidises to [1]
- A ethene
  - B ethanoic acid
  - C methane
  - D polyester
10. Addition polymerisation of ethene produces [1]
- A poly(ethene)
  - B polyester
  - C protein
  - D ethanol

## Section B · Structured questions

11 The hydrocarbons ethane, ethene and propane are compared.

(a) Give the molecular formula of each. [3]

(b) Identify the unsaturated compound. [1]

(c) Describe a chemical test and its results. [2]

(d) Define structural isomers. [2]

[Total: 8]

12 Petroleum is separated into fractions.

(a) Explain the basis of fractional distillation. [2]

(b) State two trends as chain length increases. [2]

(c) Explain why cracking is useful. [2]

(d) Write one balanced cracking equation.

[2]

**[Total: 8]**

13 Ethanol is produced by fermentation and by hydration of ethene.

(a) State conditions for fermentation.

[2]

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(b) Write the fermentation equation.

[2]

(c) State two advantages of hydration.

[2]

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(d) State two advantages of fermentation.

[2]

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**[Total: 8]**

14 Ethene reacts with bromine and can form poly(ethene).

(a) Describe the bromine reaction.

[2]

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(b) Name the reaction type.

[1]

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(c) Explain how the C=C changes during polymerisation. [2]

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(d) Describe the repeat unit, including continuation bonds and n. [3]

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**[Total: 8]**

15 Ethanoic acid reacts with ethanol.

(a) Name the organic product. [1]

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(b) Name the other product. [1]

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(c) Name the reaction type and catalyst. [2]

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(d) State two characteristic acid reactions of ethanoic acid. [2]

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(e) Explain why the reaction is reversible. [2]

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**[Total: 8]**

# Worked solutions

**Remove or fold back this section while completing the practice paper.**

## Marking routine

Mark each part separately. Award method and chemistry marks where shown, then identify the first point at which an answer becomes incorrect.

## Solutions · Multiple choice

- 1 B. Each successive chain has one extra carbon and two hydrogens.
- 2 C. Alkanes follow  $C_nH_{2n+2}$ .
- 3 C. Stronger intermolecular forces raise boiling point.
- 4 B. Large hydrocarbons split into smaller useful molecules.
- 5 B. Complete oxidation yields carbon dioxide and water.
- 6 C. The alkene adds bromine across  $C=C$ .
- 7 A. Steam adds across the double bond.
- 8 B. Yeast enzymes convert glucose anaerobically.
- 9 B. The alcohol functional group is oxidised to a carboxylic acid.
- 10 A. Ethene monomers join through opened double bonds.

## Solutions · Structured questions

11. 11(a)  $\text{C}_2\text{H}_6$ ,  $\text{C}_2\text{H}_4$  and  $\text{C}_3\text{H}_8$  [3]. (b) Ethene [1]. (c) Bromine water: ethene changes orange-brown to colourless; alkanes show no change [2]. (d) Same molecular formula but different structural formulae [2].

12. 12(a) Different boiling ranges, with vaporisation and condensation in a column [2]. (b) Boiling point and viscosity increase; volatility and flammability decrease [any 2]. (c) Converts surplus long chains to demanded fuels and alkenes [2]. (d) Any balanced alkane  $\rightarrow$  shorter alkane + alkene equation [2].

13. 13(a) Yeast, 30-40 °C and absence of oxygen [any 2]. (b)  $\text{C}_6\text{H}_{12}\text{O}_6 \rightarrow 2\text{C}_2\text{H}_5\text{OH} + 2\text{CO}_2$  [2]. (c) Fast or continuous and pure product or high yield [2]. (d) Renewable feedstock and low temperature or pressure [2].

14. 14(a) Bromine adds across the double bond and the colour disappears [2]. (b) Addition [1]. (c) The double bond opens to single bonds linking monomers [2]. (d)  $[-\text{CH}_2-\text{CH}_2-]_n$  with continuation bonds [3].

15. 15(a) Ethyl ethanoate [1]. (b) Water [1]. (c) Esterification and concentrated sulfuric acid [2]. (d) With metal, base or carbonate, giving correct products [any 2]. (e) Products can react to reform reactants and equilibrium can be established [2].

### Self-assessment

Circle the first incorrect step and classify it as knowledge, equation or formula, mole ratio, unit conversion, observation, practical method or communication.

Evidence used: Cambridge IGCSE Chemistry 0620 syllabus 2026-2028 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.